

Zisen Dai

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Education

Tsinghua University <i>Bachelor of Science in Physics GPA: 4.0/4.0 Rank: 1/28</i>	August 2023 – Present Beijing, China
UC Berkeley <i>Exchange Student in Astrophysics (Adviser: Wenbin Lu)</i>	January 2026 – Present Berkeley, CA, USA

Research Experience

Department of Astronomy, UC Berkeley <i>Research Assistant (Advised by Wenbin Lu)</i>	February 2026 – Present Berkeley, CA, USA
Tracking cutting-edge developments in high-energy transients via weekly group discussions.	
Institute for Advanced Study, Tsinghua University <i>Research Assistant (Advised by Xuening Bai)</i>	March 2025 – Present Beijing, China
- Investigating cosmic ray transport dynamics within protoplanetary disk (PPD) winds. - Developed a Python simulation framework from scratch to model particle transport via the Guiding Center Approximation, analyzing the impact of scattering mechanisms and strengths.	

Project

Rosseland-Mean Opacity Computation <i>Calculate the Rosseland-Mean Opacity of Pure Hydrogen Gas under LTE (report, code)</i>	April 2026 Berkeley, CA, USA
- Developed a Python framework to computing the Rosseland-mean opacity of pure hydrogen gas in LTE, and benchmarking it against the LANL/TOPS opacity tables. - Localized the main opacity discrepancy domains and analyzed the missing physical channels and omitted physics causing the issue.	
LBGs Selection Based on Photometric Data <i>Machine Learning-Based High-Redshift Target Selection (report, code)</i>	January 2026 Beijing, China
- Developed a LightGBM classifier on CLAUDS photometry, utilizing sample weighting and probability calibration to identify high-redshift ($z \sim 2 - 3$) Lyman Break Galaxies. - Corrected selection bias via TSR/SSR metrics, achieving 0.995 AUC and 92.0% purity — outperforming the traditional u-dropout method with a $3.3\times$ sample size increase.	
JUNO Optical Simulation & Signal Reconstruction <i>Comprehensive Physics Simulation and Machine Learning Modeling (code)</i>	August 2025 Beijing, China
- Developed a Monte Carlo Python framework to simulate complex photon transport, Fresnel reflection, and emission for 17,612 PMTs. - Engineered an ultra-high-capacity LightGBM model with multi-center Gaussian attention to reconstruct spatial-temporal Probe functions, reducing temporal integral error to $\sim 3\%$.	
Spectrum and Light Curve Analysis of Supernovae <i>Photometric and Spectroscopic Classification (report, code)</i>	June 2025 Beijing, China
- Classified SN 2025kid as a Type Ia supernova by analyzing multi-band photometric data and spectroscopic absorption features. - Extracted key parameters (peak magnitude, decline rate Δm_{15}) of SN 2025fvw via light curve fitting, utilizing the Phillips relation for standardization.	

Awards & Honors

National Scholarship <i>Highest Honor for Undergraduates, The Ministry of Education of China</i>	2024, 2025
Tsinghua Xuetaang Talents Program <i>Chi-Sun Yeh Class, Elite Physics Track</i>	2023 – 2025
Excellent Academic Performance Scholarship <i>Tsinghua University</i>	2024, 2025
Basic Disciplines Interdisciplinary Challenge <i>Third Prize, Tsinghua University</i>	2025
Chinese Physics Olympiad (CPhO) <i>First Prize in Zhejiang</i>	2022

Selected Coursework

Perfect grades (4.00/4.00) across all 23 major courses (including **6 A+**, **13 A**)

Astrophysics & Data: Statistical Methods in Astrophysics (A+), Observational Astronomy (A+), Big Data in Experimental Physics I & II (A+), Astrophysics

Advanced Physics: Statistical Mechanics (A+), Quantum Mechanics, Electrodynamics, Analytical Mechanics, Atomic and Molecular Physics

Advanced Mathematics: Mathematical Physics Equations (A+), Complex Analysis, Linear Algebra for Physical Sciences

Other Experience

Volunteer Peer Tutor <i>Drop-in Tutoring Center, Tsinghua University</i>	March 2025 – Present <i>Beijing, China</i>
• Providing one-on-one academic support and problem-solving strategies for undergraduates in advanced physics and math courses.	
Student Life Coordinator <i>Department of Organization, Zhili College</i>	March 2024 – December 2025 <i>Beijing, China</i>
• Organized college-wide student events, coordinated group activities, and facilitated team-building initiatives.	
Academic Representative <i>Physics Class 32, Zhili College</i>	September 2023 – September 2025 <i>Beijing, China</i>
• Liaised between faculty and students to facilitate academic communications and organize study groups.	